

Stats Report — Wins, Losses, Commands

Day 09

Lesson 09

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Course 02

Phase 2 — Data Lab

Time: 60–75 min

Mode: Self-study

Today's goals:

- ★ Compute win count from notes or HP
- ★ Find most-used command
- ★ Write REPORT.txt with bullet insights



Course 02 quest map: Day 09 of 18

Hook

A chart starts with sentences. Can you prove which strategy wins?

Learn



Learn

Aggregations = combining many rows into few numbers.

Examples:

- wins = rows where `dragon_hp == 0` or note contains “win”
- `fight_count` = count of `command == "fight"`
- average starting HP per round (advanced)



AI rules today

Predict/Run: AI off. **Investigate:** Socratic hints only. **Make:** you review every line.

Predict — Run — Investigate — Modify —> Make



Demo code

Folder: day09/

```
1 import csv
2 from collections import Counter
3 from pathlib import Path
4
5 rows = list(csv.DictReader(open(Path(__file__).parent / "sample_fights.csv")))
6 cmds = Counter(r["command"].strip() for r in rows)
7 wins = sum(1 for r in rows if "win" in r.get("note", "").lower())
8 print("Command counts:", dict(cmds))
9 print("Wins tagged:", wins)
```

 Try it

In terminal, `cd course02/day09`, then predict output, then run: `python demos/demo_stats.py`

 Assignment

Work in: `day09/assignment/`

A — Predict If 10 fights and dragon hits 0 HP in 4, what is win rate?

B — Practice Run demo stats on sample data.

C — Level up `assignment/stats_report.py` writes `REPORT.txt` with at least 4 numbers and 2 sentences.

D — Explain One surprise in your data.

 Checklist

- ☐ Re-read **Learn**; predict outputs (10 min)
- ☐ Assignments A–D (35–45 min)
- ☐ Run your program 3 times with different inputs
- ☐ Stuck 15 min → C-R-A-V (`../shared/prompts.md`)

Reflection

1. Easiest part today?
2. One line you can teach a friend.

Tomorrow: One bar chart with matplotlib.